

9

LENS ABERRATIONS

A wide beam of rays incident on a lens parallel to the axis, for example, is not brought to a focus at a unique point. The resulting image defect is known as *spherical aberration*. The gaussian formulas developed and used in the preceding chapters give, therefore, only an idealized account of the images produced with lenses of wide aperture.

9.1 EXPANSION OF THE SINE. FIRST-ORDER THEORY

An expansion of the sine of an angle by Maclaurin's theorem gives

$$\sin \theta = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \frac{\theta^7}{7!} + \frac{\theta^9}{9!} \cdots \quad (9a)$$

For small angles this is a rapidly converging series.

$$\sin \theta = \theta$$

When θ is small, the other angles ϕ , ϕ' , and θ' are also small, we obtain

$$\phi = \frac{r+s}{r} \theta \quad \phi' = \frac{n}{n'} \phi$$

$$\theta' = \phi' + \theta - \phi \quad s' = r - r \frac{\phi'}{\theta'}$$

equation obtained by these substitutions is none other than the gaussian formula,

$$\frac{n}{s} + \frac{n'}{s'} = \frac{n' - n}{r}$$

This equation and others developed from it form the basis of what is usually called *first-order theory*.

9.2 THIRD-ORDER THEORY OF ABERRATIONS

If all the sines of angles in the ray-tracing formulas [Eqs. (8a) to (8f)] are replaced by the first two terms of the series in Eq. (9a), the resultant equations, in whatever form they are given, represent the results of *third-order theory*. Thus $\sin \theta$ is replaced by $\theta - \theta^3/3!$, $\sin \phi$ is replaced by $\phi - \phi^3/3!$, etc. The resulting equations give a reasonably accurate account of the principal aberrations.

In this theory the aberration of any ray, i.e., its deviation from the path prescribed by the gaussian formulas, is expressed in terms of five sums, S_1 to S_5 , called the *Seidel sums*. If a lens were to be free of all defects in its ability to form images, all five of these sums would have to equal zero. No optical system can be made to satisfy all these conditions at once. Therefore it is customary to treat each sum separately, and the vanishing of certain ones corresponds to the absence of certain aberrations.

Thus, if for a given axial object point the Seidel sum $S_1 = 0$, there is no *spherical aberration* at the corresponding image point. If both $S_1 = 0$ and $S_2 = 0$, the system will also be free of *coma*. If, in addition to $S_1 = 0$ and $S_2 = 0$, the sums $S_3 = 0$ and $S_4 = 0$ as well, the images will be free of *astigmatism* and *curvature of field*.

finally S_5 could be made to vanish, there would be no *distortion* of the image. These aberrations are also known as the *five monochromatic aberrations* because they exist for any specified color and refractive index.

9.3 SPHERICAL ABERRATION OF A SINGLE SURFACE

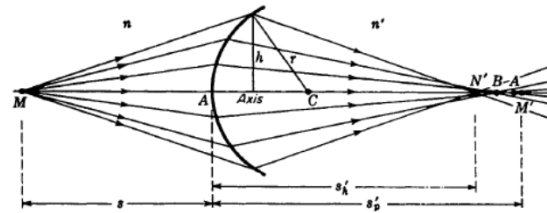


FIGURE 9B
The spherical aberration in the image of an axial object point as formed by a single spherical refracting surface.

The distance $N'M'$, as shown in the diagram, is a measure of the *longitudinal spherical aberration*, and its magnitude is found from the third-order formula

$$\frac{n}{s} + \frac{n'}{s'_h} = \frac{n' - n}{r} + \left[\frac{h^2 n^2 r}{2f' n'} \left(\frac{1}{s} + \frac{1}{r} \right)^2 \left(\frac{1}{r} + \frac{n' - n}{ns} \right) \right] \quad (9b)$$

the right-hand bracket in Eq. (9b) is a measure of the deviations from first-order theory. Its magnitude varies with the position of the object point and for any fixed point is *approximately proportional to h^2 , the square of the radius of the zone on the refracting surface through which the rays pass.*

9.4 SPHERICAL ABERRATION OF A THIN LENS

The existence of spherical aberration for a single spherical surface indicates that it may also occur in combinations of such surfaces, e.g., in a thin lens. Since many of the lenses in optical instruments are used to focus parallel incident or emergent rays, it is usual for comparison purposes to determine the spherical aberration for parallel incident light. Figure 9D(a) illustrates this special case and shows the position of the paraxial focal point F' as well as the focal points A , B , and C for zones of increasing diameter. Diagram (b) in Fig. 9D illustrates the difference between *longitudinal spherical aberration*, abbreviated Long. SA, and *lateral spherical aberration*, abbreviated Lat. SA.

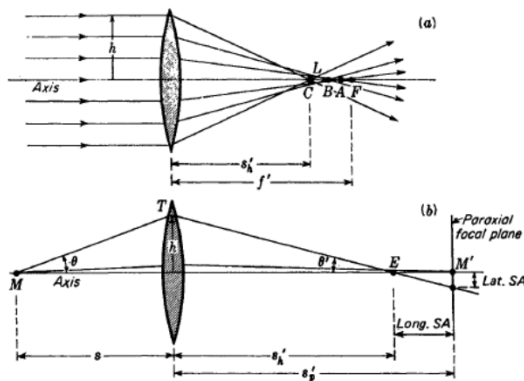


FIGURE 9D
Illustrations of lateral and longitudinal spherical aberration of a lens.

As a measure of the actual magnitudes involved in longitudinal spherical aberration, we can use the figures calculated by ray-tracing methods for some of the lenses in the preceding chapter. For example, the focal lengths for three zones of a double convex lens can be taken from Table 8B. The results are +11.52000 cm for paraxial rays, +11.11555 cm for rays from zone $h = 2.0$ cm, +9.72786 cm from zone $h = 4.0$ cm, and +6.42816 cm from zone $h = 6.0$ cm. These give a longitudinal spherical aberration of +1.79214 cm for the 4.0-cm zone, or about 15.6 percent of the paraxial focal length. A graph showing this variation of focal length with zone radius is given in Fig. 9E(a). For small h the curve approximates a parabola, and since the marginal rays intersect the axis to the left of the paraxial focal point, the spherical aberration is said to be *positive*. A similar curve, drawn for a typical double concave lens of nearly the same dimensions, is shown in Fig. 9E(b). Bending to the right, this lens is said to have *negative spherical aberration*.

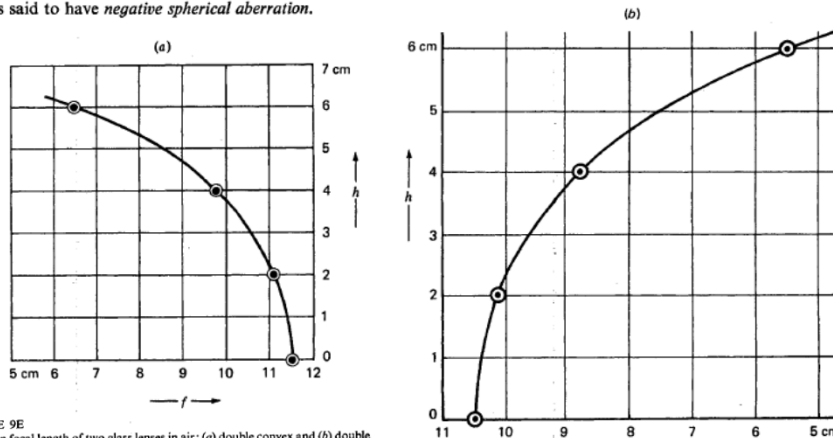


FIGURE 9E
Change in focal length of two glass lenses in air: (a) double convex and (b) double concave.

A series of positive lenses of the same diameter and paraxial focal length but of different shape is presented in Fig. 9F(a). The alteration of shape represented in this series is known as *bending* the lens. Each lens is labeled by a number q called its *shape factor*, defined by the formula

$$q = \frac{r_2 + r_1}{r_2 - r_1} \quad (9d)$$

As an example, if the two radii of a converging meniscus lens are $r_1 = -15.0$ cm and $r_2 = -5.0$ cm, it has a shape factor

$$q = \frac{-5 - 15}{-5 - (-15)} = -2$$

The usual reason for considering the bending of a lens is to find that shape for which the spherical aberration is a minimum. That such a minimum exists is shown by the graphs of Fig. 9F(b). These curves are drawn for the same lenses as shown in (a), and the values were taken from Table 9B.

Over the range of shape factors from about $q = +0.4$ to $q = +1.0$ the spherical aberration varies only slightly, since it is close to a minimum. At no point, however, does it go to zero.

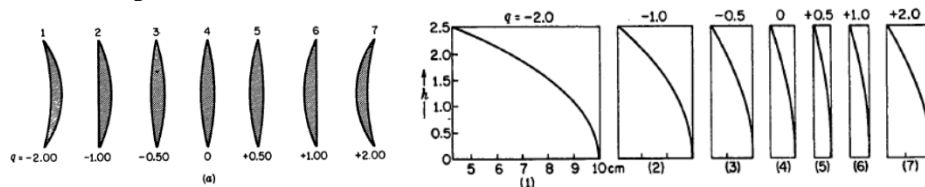


FIGURE 9F
(a) Lenses of different shapes but with the same power or focal length. The difference is one of bending. (b) Focal length versus ray height h for these lenses.

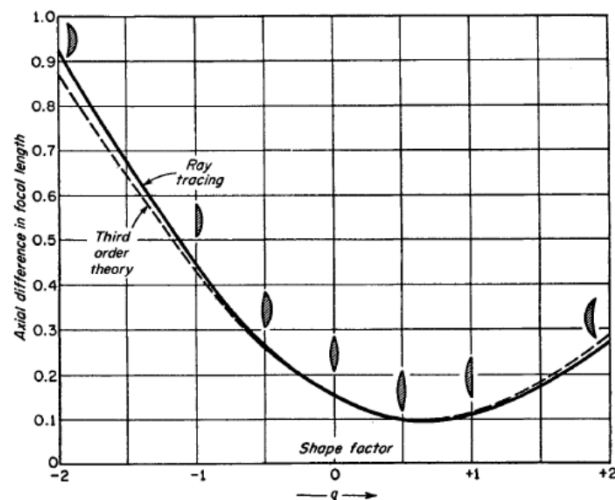


FIGURE 9G
A graph of the spherical aberration for lenses of different shape but the same focal length. For the lenses shown $h = 1$ cm, $f = +10$ cm, $d = 2$ cm, and $n' = 1.51700$.

9.7 COMA

The second of the monochromatic aberrations of third-order theory is called *coma*.

It derives its name from the cometlike appearance of the image.

Although the lens may be corrected for spherical aberration and may bring all rays to a good focus on the axis, the quality of the images of points just off the axis will not be sharp unless the lens is also corrected for coma. Figure 9I illustrates this lens defect for a single object point infinitely distant.

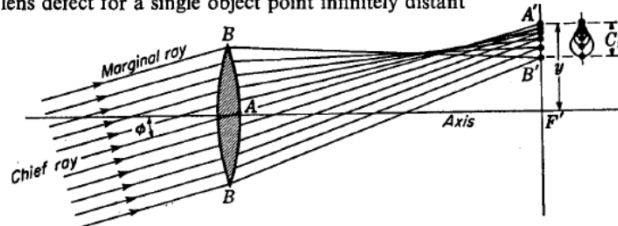


FIGURE 9I

Coma, the second of the five monochromatic aberrations of a lens. Only the tangential fan of rays is shown.

only those through the center of the lens form an image at A' . Two rays through the margin come together at B' .

If the magnification for the outer rays through a lens is greater than that for the central rays, the coma is said to be *positive*, while if the reverse is true as in the diagram, the coma is said to be *negative*.

Third-order theory shows that the radius of a comatic circle is given by

$$C_s = \frac{jh^2}{f^3} (Gp + Wq) \quad (9m)$$

where j , h , and f are the distances indicated in Fig. 9K(a) and p and q are the Coddington position and shape factors given by Eqs. (9f) and (9d). The other two constants are defined as

$$G = \frac{3(2n + 1)}{4n} \quad \text{and} \quad W = \frac{3(n + 1)}{4n(n - 1)}$$

Details of the formation of the *comatic circle* by the light from one zone of the lens are shown in Fig. 9J. Rays 1, which correspond to the *tangential rays* B in Fig. 9I, cross at 1 on the comatic circle, while rays 3, called the *sagittal rays*, cross at the top of that circle. In general all points on a comatic circle are formed by the crossing of pairs of rays passing through two diametrically opposite points of the same zone.

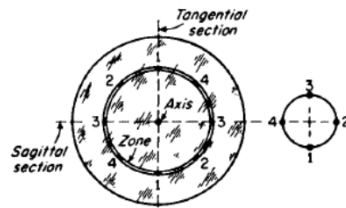


FIGURE 9J

Each zone of a lens forms a ring-shaped image called a comatic circle.

the tangential coma C_T is three times the sagittal coma C_S

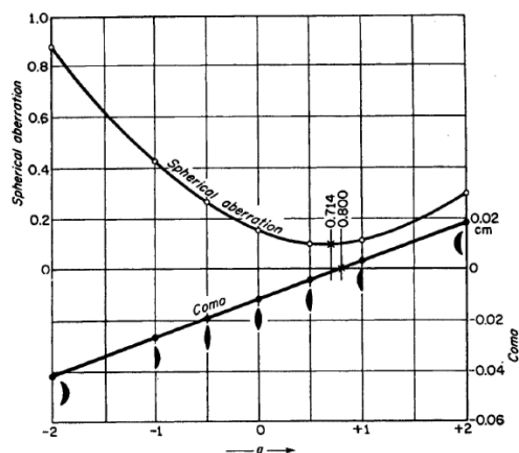
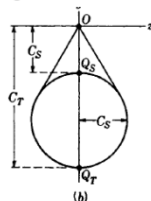


FIGURE 9L
Graphs comparing coma with longitudinal spherical aberration for a series of lenses having different shapes.

The fact that the line representing coma crosses the zero axis indicates that a single lens can be made that is entirely free of this aberration. It is important to note, for the lenses shown, that the shape factor $q = 0.800$ for no coma is so near the shape factor $q = 0.714$ for minimum spherical aberration that a single lens designed for $C_T = 0$ will have practically the minimum amount of spherical aberration.

9.6 FIFTH-ORDER SPHERICAL ABERRATION

The two curves that were given in Fig. 9G show that, for a lens having a shape factor anywhere near the optimum, the agreement between the exact results of ray tracing and the approximate results of third-order theory is remarkably good. For larger values of h , however, and for shapes further removed from the optimum appreciable differences occur. This indicates the necessity of including the fifth-order terms in the theory. The third-order equation (9e) shows that spherical aberration should be proportional to h^2 , so that the curves in Fig. 9F(b) should be parabolas. Nevertheless accurate measurements show that for larger h departures from proportionality to h^2 do occur and that spherical aberration is more closely represented by an equation of the form

$$\text{Long. SA} = ah^2 + bh^4 \quad (9i)$$

where a and b are constants. The term ah^2 represents the third-order effect and bh^4 the fifth-order effect.

Table 9C FIFTH-ORDER CORRECTION TO SPHERICAL ABERRATION
 $f = 10.0 \text{ cm}$ $r_1 = +5.0 \text{ cm}$ $r_2 = \infty$ $n = 1.500$ $d = 1.0 \text{ cm}$

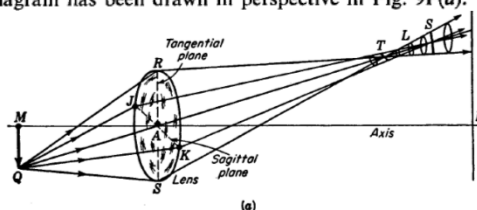
Row	h , cm	0.5	1.0	1.5	2.0	2.5	3.0
1	ah^2	0.02839	0.11356	0.25551	0.45424	0.70975	1.02204
2	bh^4	0.00011	0.00174	0.00881	0.02784	0.06797	0.14094
3	$ah^2 + bh^4$	0.02850	0.11530	0.26432	0.48208	0.77772	1.16928
4	Ray tracing	0.02897	0.11530	0.26515	0.48208	0.77973	1.16781
5	Parabola	0.02882	0.11530	0.25942	0.46120	0.71812	1.03770

9.8 APLANATIC POINTS OF A SPHERICAL SURFACE

An optical system free of both spherical aberration and coma is said to be *aplanatic*. An *aplanatic lens* can also be found for any particular pair of conjugate points, although in general it will need to be an aspherical lens. Except for a few special cases, no lens combination with spherical surfaces is completely free of both these aberrations.

9.9 ASTIGMATISM

If the first two Seidel sums vanish, all rays from points on or very close to the axis of a lens will form point images and there will be no spherical aberration or coma. When the object point lies at some distance away from the axis, however, a point image will be formed only if the third sum S_3 is zero. If the lens fails to satisfy this third condition, it is said to be afflicted with *astigmatism*, and the resulting blurred images are said to be astigmatic. The formation of real astigmatic images from a concave spherical mirror is discussed in Sec. 6.9. To help understand the formation of astigmatic images by a lens, a ray diagram has been drawn in perspective in Fig. 9P(a).



9.10 CURVATURE OF FIELD

If for an optical system the first three Seidel sums are zero, the system will form point images of point objects on as well as off the axis. Under these circumstances the images fall on the curved Petzval surface where the tangential and sagittal surfaces come together, as in Fig. 9R(b). Even though astigmatism is corrected for such a system, the focal surface is curved. If a flat screen is placed in position B , the center of the field will be in sharp focus but the edges will be quite blurred. With a screen at A , the center of the field and the field margins will be blurred, while sharp focus will be obtained about halfway out.

9.11 DISTORTION

Even if an optical system were designed so that the first four Seidel sums were zero, it could still be affected by the fifth aberration known as *distortion*. To be free of distortion a system must have uniform lateral magnification over its entire field.

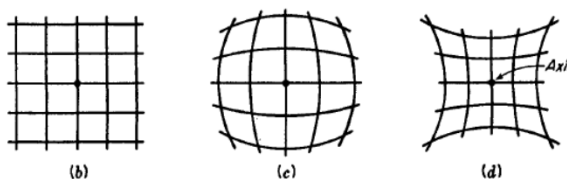


FIGURE 9T

(a) A pinhole camera shows no distortion. Images of a rectangular object screen shown with (b) no distortion, (c) barrel distortion, and (d) pincushion distortion.

9.13 CHROMATIC ABERRATION

In the discussion of the third-order theory given in the preceding sections, no account has been taken of the change of refractive index with color. The assumption that n is constant amounts to investigating the behavior of the lens for monochromatic light only. Because the refractive index of all transparent media varies with color, a single lens forms not only one image of an object but a series of images, one for each color of light present in the beam. Such a series of colored images of an infinitely distant object point on the axis of the lens is represented diagrammatically in Fig. 9X(a). The prismatic action of the lens, which increases toward its edge, is such as to cause dispersion and to bring the violet light to a focus nearest to the lens.

As a consequence of the variation of focal length of a lens with color, the lateral magnification must vary as well. This may be seen by the diagram of Fig. 9X(c), which shows only the red and violet image heights of an off-axis object point Q . The horizontal distance between the axial images is called *axial or longitudinal chromatic aberration*, while the vertical difference in height is called *lateral chromatic aberration*.

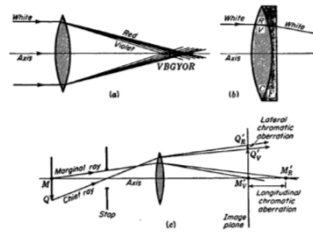


FIGURE 9X
(a) Chromatic aberration of a single lens. (b) A cemented doublet corrected for chromatic aberration. (c) Illustrating the difference between longitudinal chromatic aberration and lateral chromatic aberration.

9.14 SEPARATED DOUBLET

Another method of obtaining an achromatic system is to employ two thin lenses made of the same glass and separated by a distance equal to half the sum of their focal lengths.

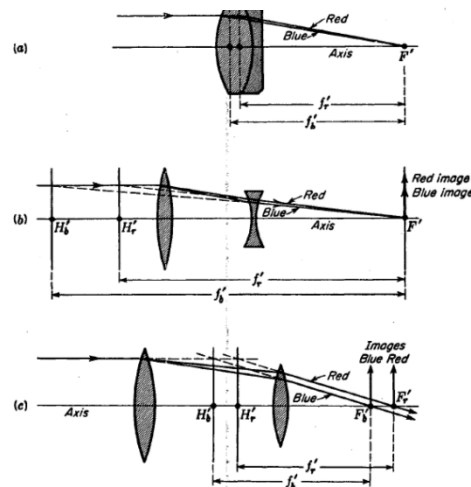


FIGURE 9Z
(a) Cemented doublet corrected for longitudinal chromatic aberration. (b) Separated doublet corrected for longitudinal chromatic aberration. (c) Separated doublet corrected for lateral chromatic aberration.